

CXS(CREDITED EXTENSIBLE STREAM)

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CHIPWEAVE TECHNOLOGIES PVT LTD

ABOUT THE CXS STREAMING INTERFACE PROTOCOL

- CXS is a high-speed, low-latency, streaming interface protocol mainly used for die-to-die communication, NoC connectivity, and chiplet architectures. It is designed to move large amounts of data efficiently between blocks, IPs, or chiplets, similar in nature to AMBA CHI streaming, AXI-Stream, or PIPE interfaces.

MAIN GOALS OF CXS

- High bandwidth with scalable data width (32 to 1024 bits)
- Low latency, minimal overhead
- Supports packetized and raw streaming data
- Supports flow control, backpressure, credits
- Designed for multi-die / chiplet interconnects
- Works with high-speed PHY (like BoW, UCle, proprietary interfaces)

MANDATORY AND OPTIONAL CXS SIGNALS

Signal Name	Width	Optional	Property	Description
CLK	1	No	—	External clock signal.
RESETn	1	No	—	External active-low reset.
CXSVALID	1	No	—	Indicates that the current cycle
CXSDATA	8 to 2048	No	CXSDATAFLITWIDTH	Data flit being transmitted (width
CXSCNTL	0, 14, 18, 22, 27, 33, 36, 44	Depends	CXSCNTLWIDTH	Contains start/end/error
CXSLAST	1	Yes	CXS_LAST	Indicates that a packet boundary
CXSPRCLTYPE	3	Yes	CXS_PROTOCOL_TYPE	Specifies the protocol type of the
CXSCRDGNT	1	No	—	Grants a single credit to the
CXSCRDRTN	1	Yes	CXSLINKCONTROL	Returns a credit to the receiver.
CXSACTIVEREQ	1	Yes	CXSLINKCONTROL	Link activation/deactivati
CXSACTIVEACK	1	Yes	CXSLINKCONTROL	Acknowledges link activation/deactivati
CXSDEACTHINT	1	Yes	CXSLINKCONTROL	Indicates receiver wishes to deactivate

CXS INTERFACE CHECKING SIGNALS

- If the CXSCHECKTYPE property is set to Odd_Byte_Parity, the interface has additional signals, which can be used to improve the integrity of the interface
- Odd_Byte_Parity describes an error detection scheme where check bits are added such that the total count of 1s across the signal and check bits is an odd number. In this scheme, signals wider than 8 bits have one bit added per byte. If the signal width is not divisible by 8, then the most significant parity bit covers less than 8 bits. For example, when CXSCNTL is 27 bits wide, CXSCNTLCHK[3] covers CXSCNTL[26:24].

COMPANY VALUES

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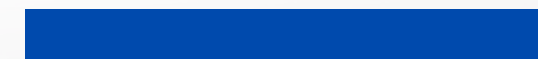
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COMPANY HISTORY

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OUR FUTURE PROJECT

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OUR COMPANY MISSION

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OUR COMPANY VISION

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OUR COMPANY PERFORMANCE

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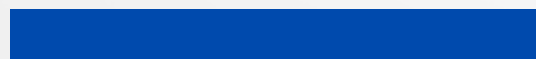
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**299K
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**112K
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MEET OUR **TEAM MEMBERS**

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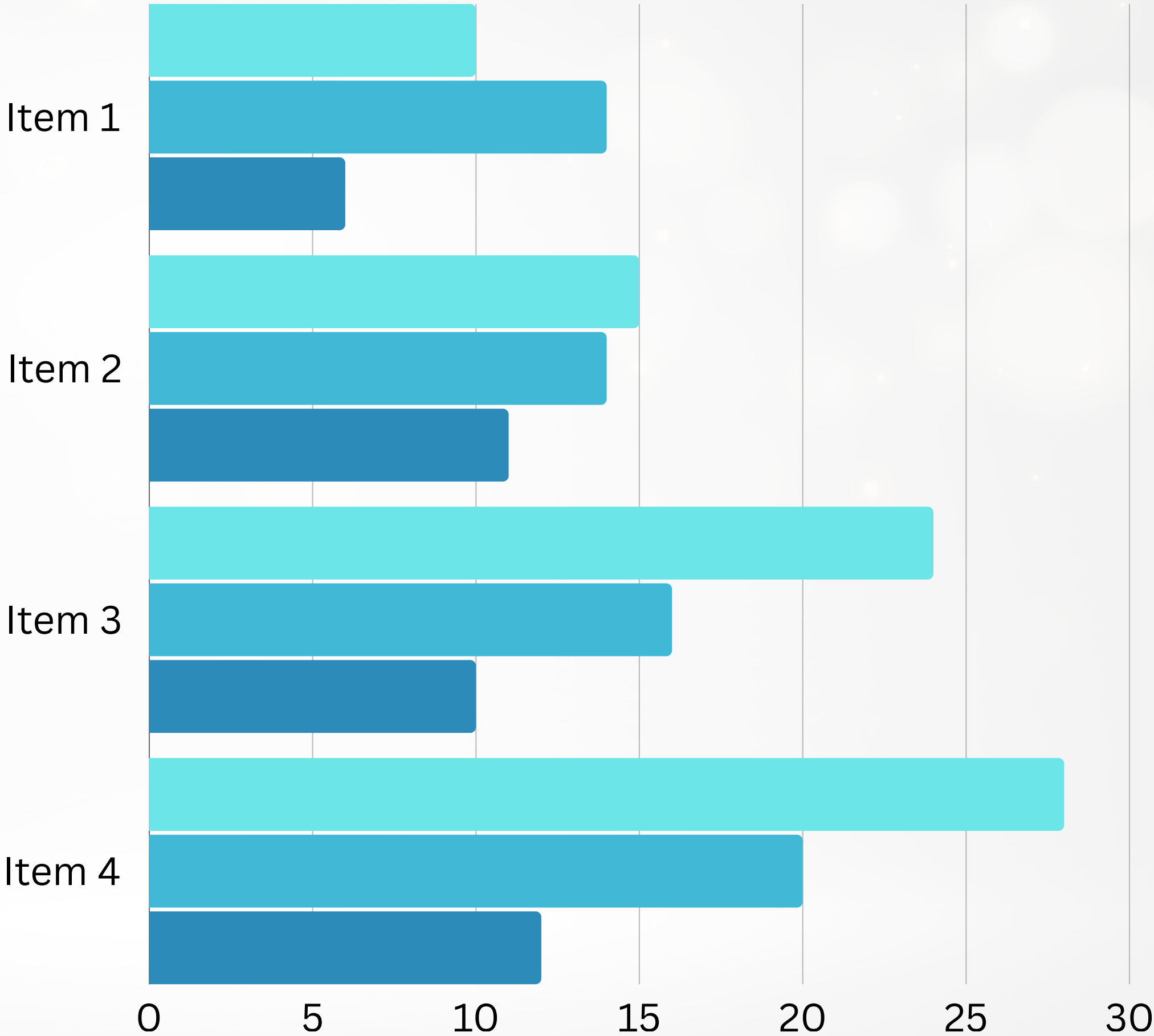


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OUR YEARLY FLOW CHART

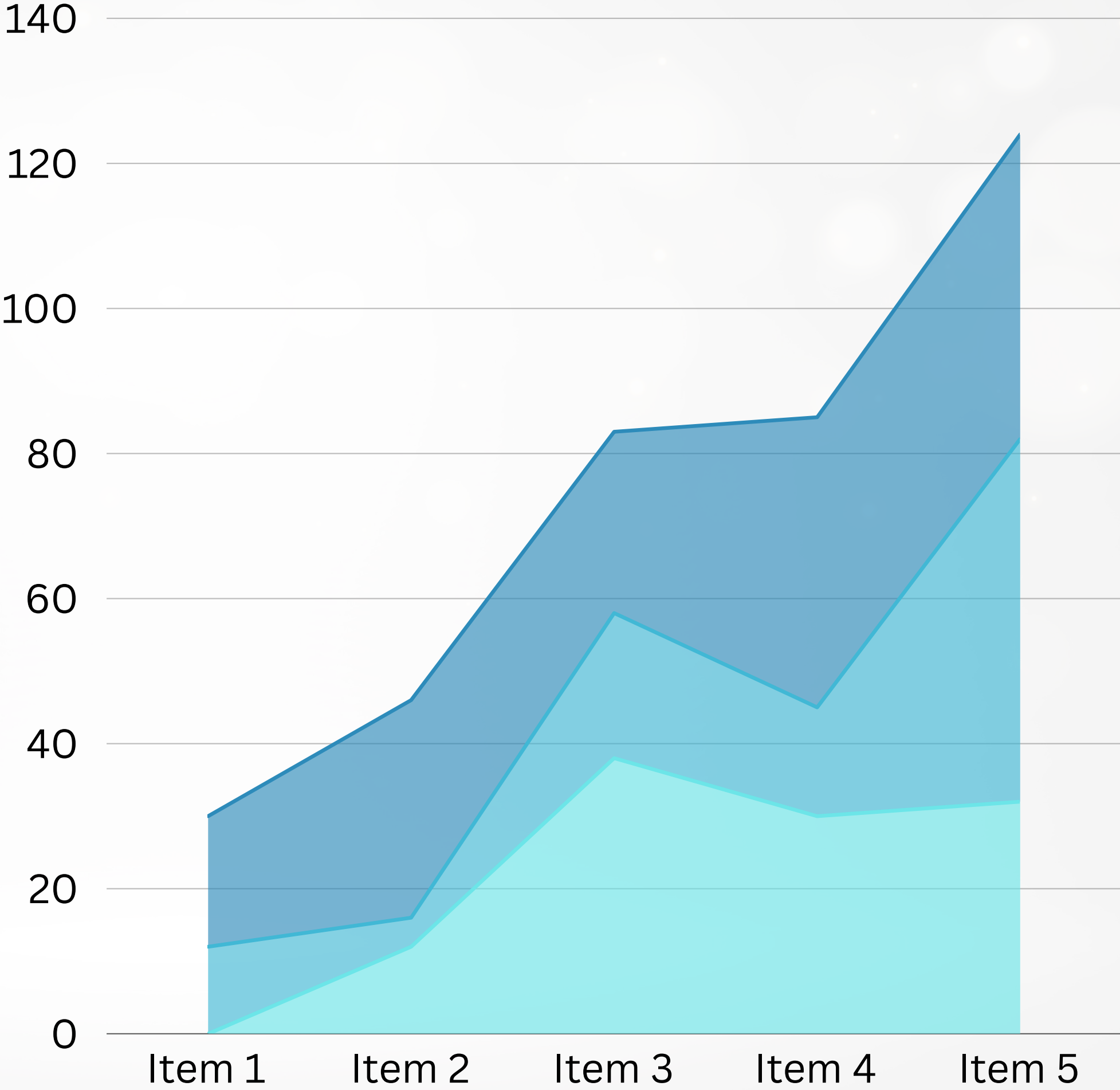
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OUR NEW PLANNING

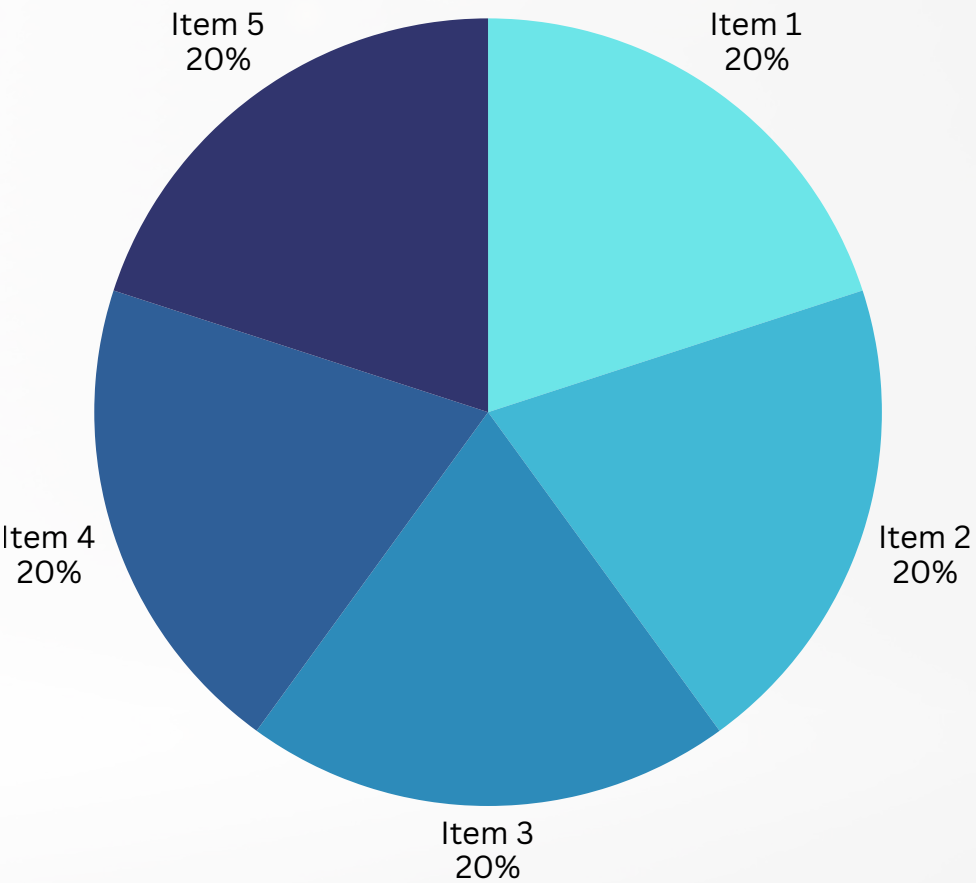
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